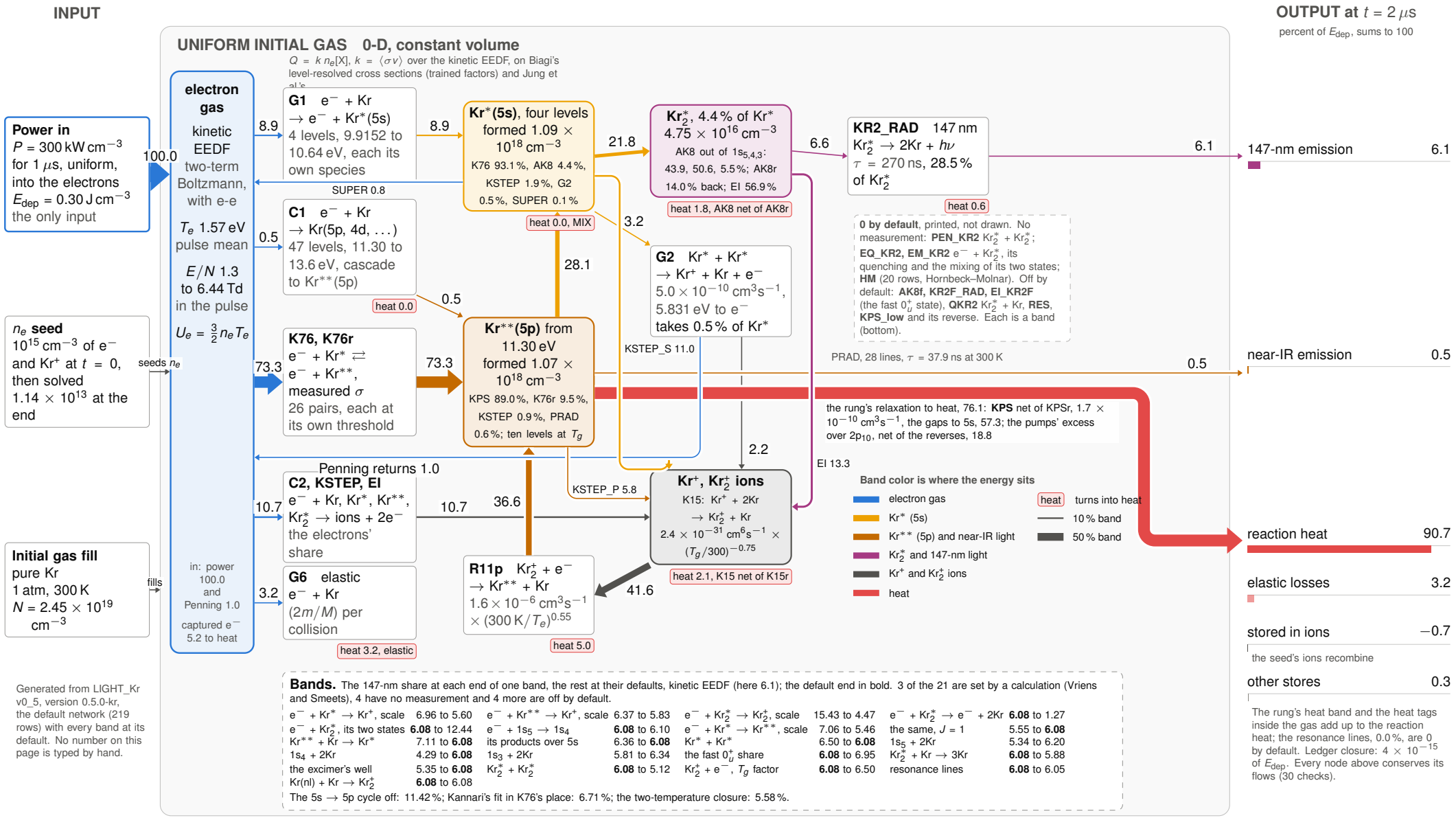
LIGHT_Kr system map, v0_5: energy flow through pure krypton

Reference point: 1 atm of Kr at 300 K, P = 300 kW cm^{-3} into the electrons for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then solved, every band at its default, run to 2 μs on the kinetic EEDF. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1. The four 5s levels are one node, Kr*, each at its own energy.



The outputs as flows, which overlap: to Kr* and Kr** by electron impact 9.4 %, to Kr+ 43.0 %, to Kr_2^* 21.4 % (formed times the 147-nm quantum), heat 90.7 %, elastic losses 3.2 %. **The 5s ↔ 5p cycle.** The electrons pump the 5s levels up to the 5p rung (K76, on Jung et al.'s measured cross sections, each pair at its own threshold) 25 times for each Kr_2^* formed net, and krypton relaxes the rung back onto every level (KPS), so the cycle is drawn as its two net bands: of the 5s atoms formed (gross), 93.1 % are pumped up; of the rung, krypton relaxes 89.0 %, the electrons return 9.5 % and ionize 0.9 %, and 0.6 % radiates. The net pumping, 9.10 × 10^{17} cm^{-3} pumps, takes 73.3 % of the input from the electrons, and the rung's relaxation turns 76.1 % into heat. The 5s atoms come from G1 (1.5 %), the rung's KPS (87.9 %) and PRAD (0.6 %), K76r (9.4 %) and the excimer's redissociation (0.6 %). **The ions**, by number: the 5s levels' ionization (KSTEP_S) makes 34 %, the excimer's (EI) 45 %, the rung's (KSTEP_P) 16 %, Penning (G2) 5 % and C2 0.089 %; they recombine onto the rung (R11p). The electrons' $1s_5 \rightleftarrows 1s_4$ mixing (EMIX) moves 0.0 % net. **The kinetic EEDF:** 338 solutions, 263 in the pulse; over its second half the field carries 1.0001 to 1.0002 of the input and the kinetic and the model's energy balances agree to 10^{-4} of it. At the end T_g = 856 K, p = 2.85 atm, n_e = 1.14 × 10^{13} cm^{-3}. **Sources** (DOIs) of the rates printed: cross sections Biagi 1999, 10.1016/S0168-9002(98)01233-9, via LXCat, 10.1016/j.chemphys.2011.04.020, times factors trained on krypton's drift velocities and ionization coefficients (5s ×0.89, 5p ×0.92, higher ×0.75, momentum transfer ×0.93 to ×1.23); K76 Jung et al. 2006, 10.1103/PhysRevA.73.022722; KSTEP and EI Vriens and Smeets 1980, 10.1103/PhysRevA.22.940; SUPER scored on Blagoev et al. 1984, 10.1088/0022-3700/17/3/013; EMIX Blagoev and Pilosov 1981, 10.1016/0375-9601(81)90875-6; AK8 Marchal et al. 2018, 10.5772/intechopen.71942, on Wieme and Lenaerts 1981, 10.1063/1.440855; AK8r LaRocque et al. 1986, 10.1063/1.450716, Killeen and Eden 1986, 10.1063/1.450794; MIX Turner 1967, 10.1103/PhysRev.158.121, Audouard et al. 1991, 10.1063/1.460834; KR2_RAD Morikawa et al. 1989, 10.1063/1.457108; G2 Leichner and Ericson 1974, 10.1103/PhysRevA.9.251, as Kannari et al. 1985 print it, 10.1063/1.334590; KPS Gazeli et al. 2021, 10.1063/5.0041471; PRAD NIST ASD, 10.18434/T4W30F; R11p Shiu and Biondi 1977, 10.1103/PhysRevA.16.1817; K15 Bhattacharya 1970, 10.1103/PhysRevA.1.1196, Grössl et al. 1981, 10.1063/1.441261, Smith et al. 1972, 10.1088/0022-3700/5/11/020; K15r and Kr_2^+'s well Signorell et al. 2001, 10.1063/1.1370939.